

Interface Capability Brief

EDI 835 Payment Integrity Demo · Stakeholder overview · 2026-08-07

1. Executive summary

Demo of opportunity idea 2.5 Payment integrity / 835 that finance can trust: ingest ERA 835s, enrich each claim payment with open-AR lineage from SQL Server, bucket matched vs exception, and emit an underpay alert CSV for shortfalls finance can act on. Demo only — not a full posting engine.

2. Who this is for

- Sources: Remit drop folder (input/inbound/*.edi) - Destinations: output/matched/, output/exceptions/, output/underpay/underpay_alerts.csv, OpenAR status updates - Demo vs production: Demo only — synthetic 835 + AR, no PM/EHR posting, no SNIP Types 1–7 for 835

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: Directory / File (DirectoryListener)
- Format / envelope: X12 835 (5010-shaped), multi-ST / multi-Transaction
- Identity fields: CLP02 patient control # → OpenAR.ClaimControlNumber
- Samples path: samples/

Integration highlights

FQCN sources: Sandbox demos (edi-835-oci-bucket, fhir-patient-exchange, hl7-healthcare-automation) + playbook. Image: pilotfish-eip:23R1.

Stage	Module / mechanism	Notes
R1 Listener	DirectoryListener	Poll \$EDI_INBOUND_DIRECTORY
R1 Format	EDITransformationModule + XPathForkingModule //Transaction	Split each ST
R1 Target processors	XPath → DatabaseSqlProcessor OpenAR lookup → XSLT score	Build RemitDecision
R1 Transport	DirectoryTransport	Stage under output/staged-decisions/
R2 Listener	DirectoryListener	Poll staged decisions
R2 Router	XPathRoutingModule accumulate=true	matched / exception / underpay CSV
R2 Transports	DirectoryTransport + NullTransport + SQL UPDATE	File buckets + AR status

4. Honest boundaries (not claimed yet)

- Add selective mock PM post in a later pass?

- Fork //CLP (+ sibling CAS context) for true multi-claim ST files?

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Poll And Score Remits” from the Web UI or documented smoke path
- Exercise route “2 - Route Remit Decisions” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- What is checked: EDI parses; CLP identity present; AR presence; paid vs expected within \$0.01 - What is NOT checked: SNIP 835, PLB full accounting, CAS semantics beyond string capture, dual-write posting to PM - Does failure block outbound? yes for unparseable EDI; no (exception theater) for business exceptions

6. State & idempotency

- Status model: OpenAR.Status OPEN → MATCHED | UNDERPAY | EXCEPTION - When state advances: after RemitDecision is built, on each routing target - Dedup keys: ClaimControlNumber + inbound file stamp (resubmits re-append CSV / overwrite decision files) - Retry / poison: bad EDI fails at EDI→XML (fail-closed); unmatched AR stays EXCEPTION

8. Dual-write / side effects

- Order: decision XML → route fan-out (files + SQL update) - Compensation: none (demo) - Demo shortcuts: underpay threshold fixed at \$0.01; CSV append without unique transaction id enforced

3. Inbound contract

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Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Poll And Score Remits: runs steps such as Poll EDI 835 Files, Fork - XPath, Extract CLP Remit Fields, Debug Write Forked Transaction.

2 - Route Remit Decisions: runs steps such as Poll Staged Remit Decisions, Extract Decision Attributes, Conditional Router, Write Matched.

7. Security & trust posture

- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: High — No PM/EHR post — Story stops at buckets + CSV
- Risk noted in design: Med — Trial X12 tables expired on 23R1 — UseInternalData=false
- Risk noted in design: Med — Claim-before-complete on AR UPDATE — File write and SQL update not XA
- Risk noted in design: Med — Multi-CLP per ST not forked per CLP — Fork is //Transaction
- Risk noted in design: Low — CSV append races — Concurrent ST

8. Seeing it work

- - Ports: SQL 14339, EIP 8110, Web UI 8111 - Volumes: ./input, ./output, ./logs, ./samples, ./documents - Heap: 1–2GB - Dependencies / cold start: SQL health + seed ~30–60s, EIP ~60–90s
- docker compose up -d --build
- - Web UI (local): <http://localhost:8111/>
- - PilotFish EIP: <http://localhost:8110/eip/>
- - Route diagrams: <http://localhost:8111/documents/route-diagrams.pdf> · <http://192.168.68.52:8111/documents/route-diagrams.pdf>
- - Capability brief: <http://localhost:8111/documents/capability-brief.pdf> · <http://192.168.68.52:8111/documents/capability-brief.pdf>
- docker compose logs -f pilotfish
- docker compose exec sqlserver /opt/mssql-tools18/bin/sqlcmd \
- -S localhost -U sa -P 'PilotFish_Demo1!' -C -d Edi835PaymentIntegrity \
- docker compose down -v

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.